

Module 4 – DBMS & SQL Theory Answers

1. SQL is a structured language used to manage and manipulate databases. It is essential for storing, retrieving and managing data.
2. DBMS manages data while RDBMS stores data in tables with relationships.
3. SQL is used to insert, update, delete and query relational data.
4. Features: Easy syntax, supports constraints, ensures data integrity, supports transactions.

SQL Syntax:

Basic components include keywords, clauses, expressions and conditions.

SELECT column FROM table WHERE condition;

Clauses define conditions like WHERE, ORDER BY, GROUP BY.

Constraints:

Constraints ensure data integrity. Types: PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK.

Primary key uniquely identifies records, foreign key creates relationships.

DDL:

DDL defines database structure. CREATE is used to create database or table.

Data types define type of data stored and constraints ensure correctness.

ALTER:

Used to modify table structure: add, modify or delete columns.

DROP:

Deletes database or table permanently.

DML:

INSERT adds data, UPDATE modifies data, DELETE removes data.

WHERE clause is important to avoid affecting all records.

DQL:

SELECT retrieves data.

ORDER BY sorts data, WHERE filters data.

DCL:

GRANT gives permissions, REVOKE removes permissions.

TCL:

COMMIT saves changes, ROLLBACK undoes changes.

JOINS:

Used to combine tables. Types: INNER, LEFT, RIGHT, FULL OUTER.

GROUP BY:

Groups records for aggregate functions like COUNT, AVG.

Stored Procedure:

Precompiled SQL code stored in database.

View:

Virtual table created from query.

Trigger:

Automatically executed on events like INSERT, UPDATE, DELETE.

PL/SQL:

Extension of SQL with programming features.

Control Structures:

IF-THEN and LOOP used for logic control.

Cursors:

Used to fetch records one by one.

SAVEPOINT:

Used to rollback to a specific point in transaction.